14-Day MERN Stack Schedule for Students

M : Mongo DB
E : Express JS
R : React JS
N : Node JS

Goal

By the end of **2 weeks**, students should be able to build and deploy a **full-stack CRUD application** with:

- **MongoDB Atlas**
- **Express.js**
- **React (Vite)**
- **Node.js**
- **REST API**
- **Basic Authentication**
- **Deployment on Vercel + Render**

Week 1 — Backend + Database

Day 1 — Node.js & Environment Setup (3 hrs)

Topics

- What is Node.js
- npm and package.json
- ES modules
- Creating a basic server

Tasks

```
mkdir mern-bootcamp
cd mern-bootcamp
npm init -y
npm install express
```

Create:

```
import express from 'express';

const app = express();

app.get('/', (req, res) => {
  res.send('MERN Bootcamp API');
});

app.listen(5000, () => console.log('Server running'));
```

Homework

- Install **Node.js**, **VS Code**, **Postman**
- Run the server successfully

Day 2 — Express.js Fundamentals (3 hrs)

Topics

- Routing
- Middleware
- JSON handling
- Request & response

Build

```
app.use(express.json());

app.get('/students', ...);
app.post('/students', ...);
app.put('/students/:id', ...);
app.delete('/students/:id', ...);
```

Mini Task

Create an **in-memory student API**.

Day 3 — MongoDB Atlas + Mongoose (3 hrs)

Topics

- MongoDB Atlas account
- Connection string
- Mongoose schemas
- Models

Install

```
npm install mongoose dotenv
```

Create

```
const studentSchema = new mongoose.Schema({
  name: String,
  course: String,
  year: Number
});
```

Outcome

Store and retrieve students from MongoDB.

Day 4 — REST API + Postman Testing (2–3 hrs)

Topics

- CRUD operations
- Status codes
- Error handling
- Async/await

Practice

Test all endpoints:

- GET
- POST
- PUT
- DELETE

Mini Project

Student Record API

Day 5 — Backend Project Day (4 hrs)

Build: Task Manager API

Features

- Add task
- View all tasks
- Update task status
- Delete task

Structure

```
backend/
├── models/
├── routes/
├── controllers/
├── server.js
└── .env
```

End of Week 1 Goal

A working **Node + Express + MongoDB backend**.

Week 2 — React + Full MERN Integration

Day 6 — React Setup with Vite (3 hrs)

Setup

```
npm create vite@latest frontend -- --template react
cd frontend
npm install
npm run dev
```

Topics

- Components
- JSX

- Props
- useState

Build

Simple **Task Card** component.

Day 7 — React Hooks + Forms (3 hrs)

Topics

- useState
- useEffect
- Controlled forms
- Form submission

Build

Add Task Form

```
const [title, setTitle] = useState('');
```

Connect form data to the console first.

Day 8 — Fetch API + Axios (3 hrs)

Install

```
npm install axios
```

Topics

- GET requests
- POST requests
- Handling loading and errors

Connect to Backend

```
axios.get('http://localhost:5000/tasks')
```

Outcome

Display tasks from MongoDB inside React.

Day 9 — Full CRUD in React (3–4 hrs)

Implement

- Create task
- Display tasks
- Edit task
- Delete task

Add

- Loading state
- Empty state
- Basic validation

This becomes the **first complete MERN CRUD app**.

Day 10 — React Router + UI Improvement (3 hrs)

Install

npm install react-router-dom

Pages

- Home
- Tasks
- Add Task
- About

Optional

Add **Tailwind CSS**:

npm install tailwindcss @tailwindcss/vite

Day 11 — JWT Authentication Basics (4 hrs)

Backend

npm install bcryptjs jsonwebtoken

Features

- Register user
- Login user
- Hash passwords
- Generate JWT token

Frontend

- Login form
- Store token in localStorage

Day 12 — Protected Routes + User Tasks (3 hrs)

Topics

- Authorization middleware
- Sending token in headers
- Protected React routes

Result

Users can only see **their own tasks**.

Day 13 — Deployment Day (3 hrs)

Backend → Render

- Push backend to GitHub
- Create Render Web Service
- Add environment variables

Frontend → Vercel
npm run build

Deploy the dist folder.

Test

- Live frontend
- Live backend
- MongoDB Atlas connection

Day 14 — Final Project + Resume Preparation (4 hrs)

Final Project

Student Task & Attendance Manager

Features

- User registration/login
- Add subjects
- Add tasks
- Mark attendance
- Dashboard with counts
- MongoDB database
- Fully deployed online

Daily Time Plan

Session	Duration	Activity
Theory	45 min	Learn new concepts
Guided Coding	60 min	Build with instructor/tutorial
Independent Practice	60 min	Rebuild without looking
Review	15 min	Commit to GitHub

Total: 3 hours/day

What Students Should Submit

After Week 1

- GitHub repo: **student-api**
- Screenshots of Postman CRUD operations

After Week 2

- GitHub repo: **mern-task-manager**
- Live **Vercel link**
- Live **Render API link**
- README.md with setup instructions

Recommended Folder Structure

```
mern-task-manager/
│
├── backend/
│   ├── controllers/
│   ├── middleware/
│   ├── models/
│   ├── routes/
│   └── server.js
│
├── frontend/
│   ├── src/
│   │   ├── components/
│   │   ├── pages/
│   │   ├── services/
│   │   └── App.jsx
│   └── vite.config.js
│
└── README.md
```

Skills Gained in 14 Days

Backend

- Node.js
- Express.js
- REST API design
- MongoDB Atlas
- Mongoose
- JWT authentication

Frontend

- React components
- Hooks
- Forms
- Routing
- API integration

Full Stack

- MERN architecture
- CRUD operations
- Authentication flow
- Deployment
- Git & GitHub workflow

Fast-Track Outcome

At the end of this **2-week schedule**, students will have:

Project: MERN Task Manager

Tech Stack

- **MongoDB Atlas** — database
- **Express.js** — backend API
- **React + Vite** — frontend
- **Node.js** — runtime
- **JWT** — authentication
- **Vercel + Render** — deployment

This is strong enough for:

- **BCA/BSc/MCA mini projects**
- **Internship applications**

- **College hackathons**
- **Portfolio demonstrations**
- **Entry-level full-stack practice**

Regards,

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